The impact of humans—who probably began visiting isolated islands in the Pacific at least 30,000 years ago—has sometimes been dramatic. In recent centuries, European people took dogs, cats, goats, and pigs with them when they colonized islands in the Pacific and elsewhere; inadvertently, they also carried rats on their boats. On many islands, rats ate the eggs of breeding seabirds and the seeds of endemic plants, some of which grew nowhere else. On the Galápagos Islands, dogs ate tortoise eggs, native iguanas, and even penguins. Goats competed with Galápagos tortoises for food and wiped out up to five species of plant on the Santiago Islands.

The arrival of humans, however, has not always reduced species richness on islands. Researchers discovered the important role of ship-assisted colonization of islands in the Caribbean. Despite its relatively small size, Trinidad, for instance, has more species of anole lizards than the much larger island of Cuba, because economic sanctions since the 1960s have meant that fewer boats (and their lizard stowaways) dock in Cuba.

“Island” habitats

In the early 1970s, American biologist James H. Brown applied the Wilson–MacArthur model to “islands” of coniferous forest on 19 mountain ridges in the Great Basin of California and Utah. The ridges are separated from each other by a vast sagebrush desert. Brown discovered that the diversity and distribution of small mammals (excluding bats) in the isolated forests could not be explained in terms of an equilibrium between colonization and extinction. Some species had become extinct, but no new species had arrived for millions of years, so Brown dubbed the mammals “relicts.” A few years later, his analysis of resident bird populations on the ridges revealed that new bird species had arrived from larger, similar forests in the Rocky Mountains to the east and in the Sierra Nevada to the west. Brown concluded that certain species groups—especially those that fly—are more likely to be successful immigrants than others.